Chapter 9
Critical Analysis of Calibration Certificate



9.1 Introduction

Sometimes, a result may fall within or outside the limit of a specification, but uncertainty can overlap with the limit, as shown in Fig. 9.1.

In Fig. 9.1, we see four cases of how a measurement result and its uncertainty may be within the limits of a specification or tolerance.

In case (a), both the result and uncertainty fall within the specified limits. This is classified as a "**compliance**."

In case (d), neither the result nor any part of the uncertainty range falls into the tolerance range. This is classified as "**non-compliance**."

In cases (b) and (c), measurements and their respective uncertainties are neither entirely within nor outside the boundaries. For this reason, these results are inconclusive and are in the **range of doubt**.

When conclusions are extracted from measurement results, measurement uncertainty should not be neglected. This is particularly important when measurements are used to verify that the result is within process tolerance or specification. Thus, we need to define an acceptance criterion for the instrument used.

This is the central theme of this chapter.

9.2 Calibration Certificate

In the International Vocabulary of Metrology (VIM—2.39), we found in Note 1 of the calibration definition: "*A calibration may be expressed by a statement, calibration function, calibration diagram, calibration curve, or calibration table. In some cases, it may consist of an additive or multiplicative correction of the indication with associated measurement uncertainty*."

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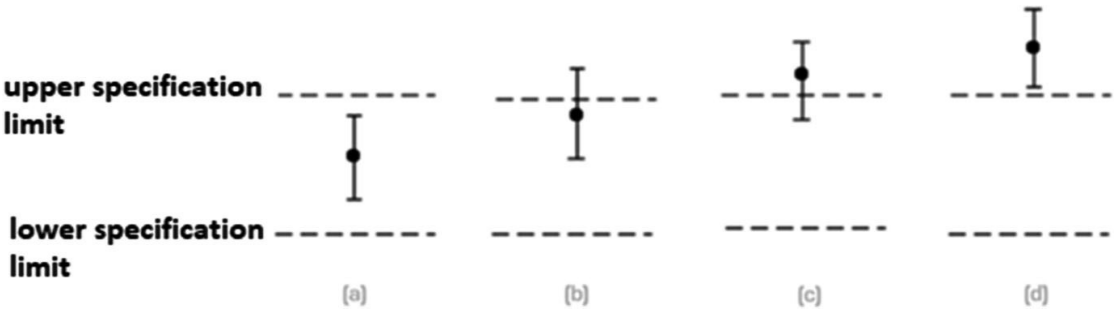


Fig. 9.1 Representation of four different measurement results within a tolerance interval. (Source: the authors)

We can consider that the document called Calibration Certificate, issued by the laboratory that performed the service, encompasses some or all expressions of calibration (declaration, function, diagram, curve, or table).

The certificate is an essential technical record and contains important information on the instrument calibration process. Through this reported information, it is possible to evaluate the conformity[1] of the measuring instrument.

The standard ISO/IEC 17025: 2017—General Requirements for the Competence of Test and Calibration Laboratories determines that the calibration results contained in a certificate must be presented clearly, objectively, and accurately, following the specific instructions of the calibration method. The certificate should also include all the information necessary for the correct interpretation of the results.

9.3 Calibration Certificate and ISO/IEC 17025

In Chap. 2, we present some management standards (ISO 9001, ISO/IEC 17025, and ISO 10012) that all highlight the importance of calibration of measurement instruments. However, none of them have a model or pattern for elaborating and presenting a calibration certificate.

ISO/IEC 17025, although it does not define the model, establishes in requirement 7.8—Results Report—what minimum information required should be included in a calibration certificate, namely:

(a) Title (e.g., calibration certificate).
(b) Name and address of the laboratory.
(c) Place of activities, including when carried out at the client's facilities or outside the laboratory's permanent facilities.

[1] According to ISO/IEC 17000: 2004, conformity assessment is any activity performed to determine, directly or indirectly, whether a product, process, system, person, or body meets relevant standards and complies with the specified requirements.

(d) Univocal identification that all components are recognized as part of a complete report and an identification of the end of the document.
(e) Client identification.
(f) Presentation of the method used in calibration.
(g) Identification of the calibrated instrument.
(h) Date of calibration.
(i) Date of issuance of the certificate.
(j) Declaration that the results apply only to the calibrated instrument.
(k) Presentation of calibration results, with their respective units of measure.
(l) Name, function, and identification of the authorized person to issue the certificate.
(m) Declaration that the certificate should only be reproduced entirely.
(n) Environmental conditions from which calibration was performed.
(o) The uncertainties of measurement.
(p) Traceability of measurements.
(q) If there is any adjustment to the instrument, the results must be reported before and after the adjustments.
(r) There should be no recommendation on the next calibration date unless the customer has previously agreed.

In Fig. 9.2, we present an example of a fictitious glass liquid thermometer calibration certificate, issued by the PP & AM Calibration Laboratory. The letters highlighted signal the necessary minimum information, as established by ISO/IEC 17025. We have not included in the certificate the information related to the letters "*q*" (adjustment) and "*r*" (new calibration recommendation).

9.4 Interpretation of Metrological Requirements in Calibration Certificates

We reiterate that every measurement instrument important to the production process should be calibrated. This allows us to know the associated errors and uncertainties. The calibration certificate provides this information.

However, a certificate does not guarantee that the instrument meets the intended requirements for its application in the measurement process, that is, a calibrated instrument is not necessarily fit for use.

From the information found in the certificate, it is necessary to evaluate this content to validate the use of the instrument.

Solved Exercise 9.1: Analytical Scale
Consider the analytical scale calibration certificate (Fig. 9.3). What should be analyzed in this certificate to evaluate the scale conformity?

Table 9.2 Maximum permissible error for scale

	Measurement range	
Class I	0 ≤ m ≤ 50 g (0 ≤ m ≤ 50,000 × 0.001 g)	50 g < m ≤ 200 g (50 < m ≤ 200,000 × 0.001 g)
Maximum permissible errors (initial verification)	± 0.5 mg (0.5 × 0.001)	± 1.0 mg (1 × 0.001)
Maximum permissible errors (in-service)	± 1.0 mg	± 2.0 mg

In the case of the scale under analysis, we can observe in the calibration certificate that the instrument is class I and the reading resolution is 0.0001 g and $e = 0.001$ g.

According Table 9.2 the maximum permissible error (initial and in-service) per measuring range will be:.

In the calibration certificate, we find the scale measurement bias. Comparing the measurement bias values indicated on the certificate with the maximum allowable error table by range, we can conclude that:

- $m \leq 50$ g: scale bias (0.8 mg) is lower than MPE in-service (1.0 mg) ⇨ APPROVED.
- 50 g $< m \leq 150$ g: scale bias (0.9 mg) is lower than MPE in-service (2.0 mg) ⇨ APPROVED.
- 150 g $< m \leq 200$ g: scale bias (3.3 mg) is higher than MPE in-service (2.0 mg) ⇨ DISAPPROVED!

Conclusion The scale needs to be adjusted to reduce its bias at the end of the range, but if the adjustment is not performed, it should only be used in measurements between 0 and 150 g.

Second: Final Uncertainty

We consider whether to weigh a particular substance in the laboratory using our calibrated scale. We also think that the mass value of this product must be understood in the interval of (50.0000 ± 0.0100) g, that is, it must be between 49.9900 g (T_L—the lower limit of tolerance) and 50.0100 g (T_U—the upper tolerance limit).

The decision rule to be adopted is to consider a false positive of 2.5 % or less.

According to Table 8.11 (Chap. 8), if we use a guard band equal to expanded uncertainty, we will have an acceptance interval where the probability of a value is not equal to or less than 2.5%. The scale sheet certificate states that the measurement uncertainty value (U) of 50 g is 0.9 mg (Fig. 9.4).

So, $T_L = 49.9900$ g and $T_U = 50.0100$ g. The control limits (A_L and A_U) are:

$$A_L = T_L + U = 49.9900 + 0.0009 = 49.9909 \text{ g}$$

$$A_U = T_U - U = 50.0100 - 0.0009 = 50.0091 \text{ g}$$

According to the decision rule adopted, the mass result will always be acceptable if it falls within the acceptance interval of 49.9909 g to 50.0091 g.

$$MPE_{mass} \leq \frac{MPE_{balance}}{3}$$

$$MPE_{mass} \leq \frac{1\ mg}{3} = 0.333\ mg$$

Over 50 g:

$$MPE_{mass} \leq \frac{2\ mg}{3} = 0.666\ mg$$

Referring to Table 7.1 (Chap. 7):

- Up to 50 g—Class F_1 mass.
- Between 50 g and 200 g—Class E_2 mass.

Suppose the bias presented in the mass calibration certificate to be used in the scale verification is greater than the maximum permissible error of the class. In that case, the mass changes to a lower class.

9.5 Acceptance Criterion (AC) of a Measurement Instrument

To date, we have discussed the analysis of a calibration certificate by verifying if it is fit for use and meets the requirements of technical regulation, as well as calculating the upper (A_U) and lower (A_L) control limits.

In this section, we will analyze a calibration certificate by verifying whether the measurement instrument is approved for use in an industrial measurement process.

The first step is to know the acceptance criterion adopted by the industry. The acceptance criterion considers the maximum uncertainty of the instrument used to verify the product's conformity with its specification. Since the acceptance criterion is a fraction of tolerance, it will always be less than the process tolerance.

To determine the *AC*, we must adopt the following relationship:

$$AC = \frac{TL}{TUR} \tag{9.1}$$

TUR ranges from 3 to 10, and *TL* is the process tolerance interval.

In Solved Exercise 9.1, the tolerance interval (*TL*) is equal to 0.01 g for measurement at point 50 g.

If we adopt:

(a) $TUR = 10 \rightarrow AC = 0.001$ g $\rightarrow$ $A_L = 49.9900 + 0.001 = 49.9910$ g and. $A_U = 50.0100–0.001 = 50.0090$ g.

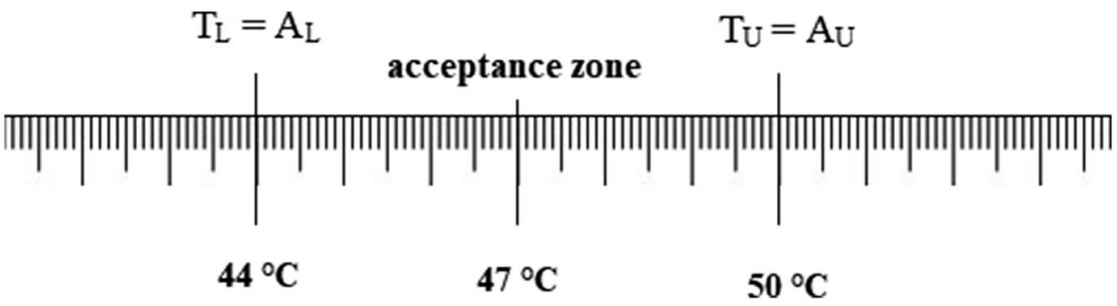


Fig. 9.6 Acceptance zone

(g) Note that the thermometer calibration certificate of Fig. 9.6 has its maximum uncertainty (U_{maximum}) in the use range (44–50) °C of 0.4 °C (Fig. 9.7).

$$B = 0.1\,^{\circ}C; U = 0.3\,^{\circ}C$$

$$0.1 + 0.3 = 0.4\,^{\circ}C$$

$$U\text{maximum}\ (0.4\,^{\circ}C) < AC\ (1\,^{\circ}C)$$

Thermometer approved for use in the process!

9.6 Proposed Exercises

9.5.1 Regarding the Calibration Certificate, check (*R*) for the correct statements and (*W*) for the wrong ones (Table 9.13).

9.5.2 Table 9.14 shows the calibration result of a Bourdon-type gauge, class 1.0, with a measurement range between 0 and 60 bar and 0.5 bar resolution.

Based Table 9.14, ensure the object gauge can continue to be used as class 1.0.

9.5.3 Consider the calibration certificate presented. According to ISO/IEC 17025: 2017, what information is needed and missing? (Fig. 9.8)

9.5.4 A 25 mL burette, class B, was calibrated, and the data obtained were: measured volume = 25.05 mL; uncertainty = 0.01 mL. Based on this information, ensure the burette can continue to be used as class B and meets an AC of 0.1 mL.

9.5.5 The pulley of an engine must have a width of (25.000 ± 0.012) mm and a diameter of (960.0 ± 1.5) mm. To measure the width, a micrometer with a maximum uncertainty of 0.002 mm and a digital tape with a maximum uncertainty of 0.5 mm for the diameter. What should be the control limits of the manufacture of this pulley considering these measuring instruments? Adopt a 95 % decision rule.